

BAY CITY, TX, USA

WMO: 720273

Lat: 28.973N

Lon: 95.863W

Elev: 45

StdP: 14.67

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.1	(c) 35.5	(d) 18.4	(e) 13.9	(f) 41.9	(g) 23.4	(h) 17.7	(i) 47.2	(j) 24.5	(k) 59.4	(l) 21.9	(m) 59.3	(n) 5.4	(o) 350	(p) 0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 17.6	(c) 96.2	(d) 78.4	(e) 94.6	(f) 78.5	(g) 92.9	(h) 78.6	(i) 82.3	(j) 88.3	(k) 81.5	(l) 87.9	(m) 80.8	(n) 87.3	(o) 9.9	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.2	162.2	84.1	80.2	156.9	83.6	79.3	152.4	83.2	46.0	88.4	45.1	88.2	44.3	87.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.8	(b) 18.7	(c) 16.9	(e) 25.7	(f) 99.9	(g) 5.5	(h) 3.0	(i) 21.7	(j) 102.1	(k) 18.5	(l) 103.8	(m) 15.4	(n) 105.5	(o) 11.4	(p) 107.7		
(a) 20.8	(b) 18.7	(c) 16.9	(e) 23.9	(f) 83.3	(g) 4.9	(h) 2.4	(i) 20.4	(j) 85.0	(k) 17.5	(l) 86.5	(m) 14.7	(n) 87.8	(o) 11.1	(p) 89.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.9	54.6	59.2	64.8	70.2	76.9	82.6	83.5	84.0	80.3	73.0	63.7	57.3
	DBStd	12.32	9.02	9.46	7.92	6.33	4.72	2.70	2.68	3.08	3.80	6.85	8.98	9.39
	HDD50	133	54	28	6	1	0	0	0	0	0	0	8	36
	HDD65	1101	341	203	104	26	2	0	0	0	0	18	132	275
	CDD50	7760	195	286	466	607	833	977	1038	1054	909	713	420	262
	CDD65	3253	18	41	99	182	369	527	573	589	459	266	94	36
	CDH74	31873	56	128	399	1166	3272	5944	6581	6811	4369	2378	609	160
	CDH80	13131	1	4	26	169	999	2692	3112	3334	1874	809	105	6
(14) Wind	WSAvg	7.0	7.8	8.5	8.7	8.9	8.1	6.7	5.4	5.1	5.2	6.1	6.8	7.4
(15) Precipitation	PrecAvg	47.2	3.9	2.5	3.1	3.3	3.8	4.7	4.6	3.4	5.7	4.4	4.3	3.5
	PrecMax	65.9	10.7	8.5	9.3	8.3	8.3	15.4	18.3	8.1	16.9	11.3	11.7	8.3
	PrecMin	26.3	0.1	0.4	0.6	0.2	0.0	0.6	0.6	0.7	1.2	0.0	0.4	0.9
	PrecStd	11.4	2.7	2.1	2.1	2.3	2.5	3.3	4.3	2.2	4.3	3.1	3.0	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	80.5	82.4	86.9	91.2	98.1	96.9	99.1	97.1	91.1	86.2	80.9
		MCWB	65.3	67.5	69.5	70.9	75.2	76.8	78.5	77.8	78.4	77.5	74.2	71.3
	2%	DB	75.4	77.4	80.3	83.6	89.0	94.0	94.9	96.7	93.7	89.1	83.0	78.3
		MCWB	67.0	68.4	70.0	72.3	75.8	78.0	78.4	78.6	77.5	76.7	72.7	70.6
	5%	DB	72.7	75.3	78.4	81.9	87.2	92.1	93.4	95.0	91.2	87.2	80.6	75.4
		MCWB	66.5	67.7	69.6	71.9	75.6	78.5	78.8	79.2	77.8	75.4	71.5	70.0
	10%	DB	70.2	72.8	76.4	80.1	85.7	90.5	91.8	93.0	89.6	85.0	77.6	72.8
		MCWB	64.9	66.9	68.6	71.2	75.3	78.4	78.8	79.2	77.4	74.0	69.9	68.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	73.4	74.2	77.7	79.8	82.8	82.8	83.5	82.4	81.5	77.6	75.3
		MCDB	73.9	76.4	78.1	81.2	84.2	88.5	87.6	89.2	89.0	86.1	81.2	77.0
	2%	WB	70.7	72.1	73.1	76.5	78.7	81.4	81.9	82.4	81.3	79.9	75.9	73.4
		MCDB	72.7	75.0	76.4	79.9	83.8	87.0	87.7	89.0	87.9	84.6	79.4	76.1
	5%	WB	68.9	71.0	72.1	75.3	78.1	80.6	81.1	81.5	80.3	78.3	74.4	71.9
		MCDB	71.3	73.6	75.4	78.8	83.2	86.7	87.5	88.7	86.9	83.4	77.9	74.6
	10%	WB	66.5	69.4	71.0	74.2	77.3	79.9	80.3	80.5	79.4	77.0	72.6	69.7
		MCDB	69.4	71.6	74.4	77.8	82.5	86.4	87.0	87.8	85.8	82.2	76.4	71.7
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	18.3	18.1	17.9	16.3	16.0	16.2	17.6	17.6	20.5	20.4	19.1
		MCDBR	21.6	17.7	16.8	17.2	16.7	18.3	18.3	19.9	19.8	19.7	20.3	19.7
		MCWBR	14.8	11.4	8.8	8.8	6.3	5.8	5.4	5.7	7.1	8.5	11.4	13.8
	5% WB	MCDBR	17.8	14.7	13.6	13.0	13.6	15.1	15.8	17.8	16.5	16.2	16.5	18.0
		MCWBR	15.1	11.0	8.0	7.6	6.0	5.7	5.6	6.4	6.8	8.7	11.5	14.7
(40) Clear-Sky Solar Irradiance	taub		0.350	0.363	0.381	0.439	0.489	0.462	0.489	0.474	0.454	0.387	0.379	0.354
	taud		2.487	2.460	2.398	2.235	2.126	2.267	2.207	2.251	2.276	2.452	2.420	2.483
	Ebn at Noon		281	287	288	273	259	264	257	260	262	275	268	274
	Edh at Noon		30	33	37	45	50	43	46	44	41	33	31	28
(44) All-Sky Solar Radiation	RadAvg		921	1060	1434	1713	1930	2054	1992	1905	1614	1410	1060	827
	RadStd		96	174	127	141	129	140	184	125	158	151	113	101

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+178

Nomenclature: See separate page